University of BRISTOL

Lecture 10
BVP solutions

ASDA/Dynamics of Rotor Systems

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This lecture

- BVP solution approach 1: Analytical solution
  - manual derivation
- BVP solution approach 2: Approximate solution
  - Collocation method
    - BVP4C: collocation solver in Matlab
- Example implementation and comparative study

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Example analytical solution

A non-rotating hingeless blade is modeled as a cantilevered beam. The distributed blade properties are: EI_F=5000 kg$*$m^2, m=9 kg/m and the rotor radius is R=6.4 m. The rotor speed is Ω=0 rad/s. Solve analytically the corresponding eigenvalue BVP.

Consider the homogeneous flapping blade EoM at Ω=0 rad/s

$$\left(EI_F(x)W''(x)\right)'' - \omega_F^2 m(x)W(x) = 0$$

Note that the system, when moving from the PDE to ODE, is assumed to vibrate at the *unknown* frequency ω_F.

The blade properties are constants

$$EI_F W^{iv}(x) - \omega_F^2 m W(x) = 0 \;\Rightarrow\; \underline{W^{iv}(x) - \beta^4 W(x) = 0},\; \underline{\beta^4 = \omega_F^2 m / EI_F}$$

The trial solutions are assumed in the form W(x)=Ae^{rx}

$$W^{iv}(x) - \beta^4 W(x) = 0 \;\Rightarrow\; (r^4 - \beta^4)Ae^{rx} = 0 \;\Rightarrow\; \underline{r^4 - \beta^4 = 0} \qquad \begin{cases} r_{1,2} = \pm\beta \\ r_{3,4} = \pm i\beta \end{cases}$$

The total solution is

$$W(x) = B_1 e^{\beta x} + B_2 e^{-\beta x} + B_3 e^{i\beta x} + B_4 e^{-i\beta x}$$

$$W(x) = A_1\cosh\beta x + A_2\sinh\beta x + A_3\cos\beta x + A_4\sin\beta x$$

$$\begin{cases} e^{\pm\beta x} = \cosh\beta x \pm i\sinh\beta x \\ e^{\pm i\beta x} = \cos\beta x \pm i\sin\beta x \end{cases}$$

The superposition of all possible (four) solutions constitutes the full (free) vibration solution.

Example analytical solution, continued …

The total solution contains *four* unknown constants A_i

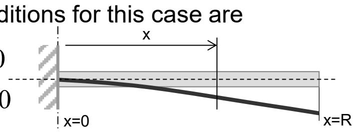
$$W(x) = A_1\cosh\beta x + A_2\sinh\beta x + A_3\cos\beta x + A_4\sin\beta x$$

The constants are determined with the help of sufficient number of boundary conditions (BCs).

The boundary conditions for this case are

$$x = 0,\; W(0) = 0 \qquad x = R,\; W''(R) = 0$$

$$x = 0,\; W'(0) = 0 \qquad x = R,\; W'''(R) = 0$$



All four BCs, one after another, are substituted into W(x) to give

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & \beta & 0 & \beta \\ \beta^2\cosh\beta R & \beta^2\sinh\beta R & -\beta^2\cos\beta R & -\beta^2\sin\beta R \\ \beta^3\sinh\beta R & \beta^3\cosh\beta R & \beta^3\sin\beta R & -\beta^3\cos\beta R \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \;\Rightarrow\; \mathbf{B}\mathbf{x} = \mathbf{0}$$

This is a homogeneous system of four linear equations in four unknowns. The only case when it has non-trivial solutions arises when the system matrix is *singular*.

To achieve the non-trivial (non-zero) solution

$$\det(\mathbf{B}) = 0 \;\Rightarrow\; \cos\beta R\cosh\beta R + 1 = 0 \qquad \rightarrow \underline{\beta_j R = 1.88, 4.69, 7.85, (n-1/2)\pi}$$

This is the *frequency equation*. This equation is solved numerically. Theoretically, it has an infinite number of solutions.

Example analytical solution, continued …

The solutions $\beta_j R$ are then used to determine the *natural frequencies*:

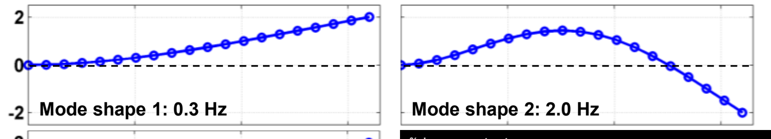
$$\beta^4 = \omega_F^2 \frac{m}{EI_F} \quad \Rightarrow \quad \omega_F = \frac{(\beta_j R)^2}{R^2}\sqrt{\frac{EI_F}{m}}$$

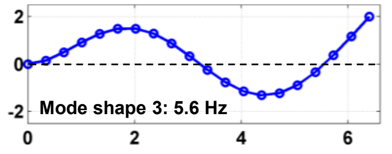
j-th numerical solution βjR of the frequency equation gives the *j*-th natural frequency. An infinite number of solutions implies an infinite number of natural frequencies (and corresponding mode shapes).

The constants A_i are found by back-substitution of $\beta_j R$ to **Bx**=**0** and solving for **x**. The resulting $W_j(x)$ is called the j-th mode shape (function):

$$W_j(x) = \cosh\beta_j x - \cos\beta_j x - \xi_j(\sinh\beta_j x - \sin\beta_j x),\ \ \xi_j = (\cosh\beta_j R + \cos\beta_j R)/(\sinh\beta_j R + \sin\beta_j R)$$

Plot the typical mode shapes for this case using Matlab:






```
% beam constants
n=4:6; bl=[1.88,4.69,7.85,(n-1/2)*pi];
kn=(cosh(bl)+cos(bl))./(sinh(bl)+sin(bl));
% natural frequency
omega=@(m,EI,L)((bl./L).^2*sqrt(EI./m));
% mode shape
Wn=@(x,L,ii)(cosh((bl(ii)/L).*x)-cos((bl(ii)/L).*x)-...
   kn(ii).*(sinh((bl(ii)/L).*x)-sin((bl(ii)/L).*x)));
```


Method of Weighted Residuals

To solve BVPs, we usually *discretize* the problem and find the corresponding *approximate solution*. Wide range of approximate methods exist: Finite Element Method, Finite Differences, etc. We use the Method of Weighted Residuals (MWR).

Symbolically, we can write our problems: HW=L

e.g. $(EI_F(\circ)^{iv} - \omega_F^2 m)W = 0,\ HW = 0$

Depending on W(x), we have:

- $\boxed{HW(x) - L = 0}$ *W*(x) is a solution
- $\boxed{HW(x) - L \approx 0}$ *W*(x) is ***almost*** a solution
- $\boxed{HW(x) - L \neq 0}$ *W*(x) is not a solution

We seek, numerically, the solutions where *R*(*W*)=*HW-L* is small everywhere in *D*. *R*(*W*) is the *residual* between RHS and LHS. MWR aims to achieve *R*(*W*)→0 in two steps:

(1) Choose the *parameterised* trial solution: $\tilde{W}(x) = \sum_{i=1}^{N} a_i u_i(x) = a_1 u_1(x) + a_2 u_2(x) + \ldots$

$u_i(x)$ are the *known* and suitably chosen *N trial* functions which comply with the BCs and are "well-behaved" (independent in some sense, etc.), a_i are the *unknown* and *solved* real valued coefficients.

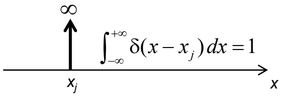
(2) Solve for a_i such that *R*(x)→0 by requiring: $\int_D R(\tilde{W}(x)) h_j(x)\,dx = 0,\ \ j = 1,\ldots,N$

$h_j(x)$ are the *known* and *suitably* chosen *N weighting* functions that, again, comply with some BCs and some other conditions.

Collocation method

A particular choice of $h_j(x)$ leads to different versions of MWR such as Galerkin method, Least Square method, **Collocation method**, etc.

Collocation method uses $h_j(x)=\delta(x\text{-}x_j)$ (i.e. shifted Dirac delta function):



Then, the conditions for the residuals are as follows:

$$\int_D R(\tilde{W}(x))\delta(x-x_j)\,dx = R(\tilde{W}(x_j)) = 0$$

The residuals are required to be 0 only in the set of N chosen collocation points x_j.

Further, our BVPs are linear in $W(x)$. Then, the collocation equations are

$$H\tilde{W}(x_j) - L = 0 \quad \rightarrow \quad \sum_{i=1}^{N} a_i\, Hu_i(x_j) - L = 0$$

This is a system of N linear equations for N chosen x_j collocation points with N unknown a_i coefficients.

Basic collocation example

Use collocation approach and method to discretize the eigenvalue BVP for a non-rotating flapping blade with *variable sectional properties*.

General field equation after differentiation:

$$EI\,W^{iv} + 2EI'W''' + EI''W'' - \omega^2 mW = 0$$

The residual function and the trial solution are:

$$R(W(x)) = EI\,W^{iv} + 2EI'W''' + EI''W'' - \omega^2 mW \qquad \tilde{W}(x) = \sum_{i=1}^{N} a_i u_i(x)$$

u_i are chosen to be the known mode shapes which comply with this problem's BCs (e.g. those of the non-rotating system, solved before, with the "mean"-constant sectional properties).

Resulting weighted residual equations are:

$$\int_D R(\tilde{W}(x))\underline{\delta(x-x_j)}\,dx = R(\tilde{W}(x_j)) = 0$$

$$\sum_{i=1}^{N} a_i\left(EI_j\,u_i^{iv}(x_j) + 2EI'_j\,u_i'''(x_j) + EI''_j\,u_i''(x_j)\right) = \omega^2 \sum_{i=1}^{N} a_i\, m_j u_i(x_j)$$

The above equation is written for all collocation (discretisation) points.

The above system of collocation equations constitutes an *algebraic eigenvalue* problem which can be expressed in the following matrix form:

$$\mathbf{Ka} = \lambda\mathbf{Ma},\ \mathbf{a} = [a_i]$$

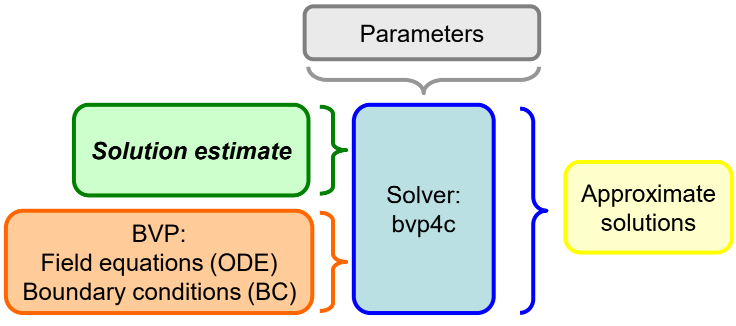
$$\mathbf{K} = [k_{ji}] = [EI_j\,u_i^{iv}(x_j) + 2EI'_j\,u_i'''(x_j) + EI''_j\,u_i''(x_j)],\ \mathbf{M} = [m_{ji}] = [m_j u_i(x_j)]$$

Collocation method using bvp4c

To avoid the need to develop our own numerically-robust collocation scheme, we rely on existing and proven numerical solvers. In this case, we use:

bvp4c[1] implements a *collocation method* for BVPs $\mathbf{y}'=\mathbf{f}(x,\mathbf{y},\mathbf{p})$, $a \le x \le b$, subject to nonlinear, two-point boundary conditions $\mathbf{g}(\mathbf{y}(a),\mathbf{y}(b),p)=\mathbf{0}$. The continuous *approximate solution* $\mathbf{S}(x)$ is a *cubic polynomial* on each subinterval $[x_n,x_{n+1}]$ of a mesh $a=x_0<x_1<\ldots<x_N=b$. It *satisfies* (**collocates**) the BCs $\mathbf{g}(\mathbf{S}(a),\mathbf{S}(b)) = \mathbf{0}$ and the field equations at both ends and the mid-point of each subinterval.

Constituent elements of bvp4c:



[1] Shampine, Kierzenka, Reichelt, Solving Boundary Value Problems for Ordinary Differential Equations in Matlab with *bvp4c*, 2000.

Annotated bvp4c example

Problem parameters

Solver parameters and pre-allocation

Repeated execution of the solver at the range of excitation frequencies

Plotting resuts

Field equation for the rotating flapping blade, L=0, m,EI=const

Rigid hub, unit and harmonic vertical tip excitation

Initial solution estimates for each solver execution

```
function example
% Parameters
E=7e10; I=2.9268e-09; m=0.320; R=2.3; EI=E*I;
Omega=2*pi*4; F0=1;

xi=linspace(0,R,10);
nw=200;omegavec=linspace(0,2*pi*80,nw);
Y=zeros(1,nw);

for isol=1:nw
   omega=omegavec(isol);
   solinit=bvpinit(xi,@modelini);
   sol=bvp4c(@fieldeq,@modelbc,solinit);
   sol0=sol; Y(isol)=sol.y(1,end);
end

% plotting results
figure, plot(omegavec/2/pi,log10(abs(Y)),'.-')
title('Response'), xlabel('frequency [Hz]'), ylabel('|Y|')
% ------------------------------------------------------------
   function dydx = fieldeq(x,y)
      % Field equation: EI*W''''-T'*W'-T*W''-w^2*m*W=0
      omega_term=Omega^2*m*((R^2-x^2)*y(3)/2-x*y(2));
      dydx=[
         y(2); y(3); y(4); ((omega^2)*m*y(1)+omega_term)/EI ];
   end
% ------------------------------------------------------------
   function res = modelbc(ya,yb)
      % Boundary conditions: W(0)=W'(0)=W''(R)=0,EI*W'''(R)=F0
      res=[
         ya(1); ya(2); yb(3); EI*yb(4)-F0];
   end
% ------------------------------------------------------------
   function y0=modelini(x)
      % Initial estimate
      if isol==1
         y0=zeros(4,1);
      else
         y0=interp1(sol0.x,sol0.y.',x,'linear','extrap').';
      end
   end
end
```

Unit excitation force amplitude, thus the normalized steady state harmonic responses (i.e. FRF)

Explored excitation frequency range and individual frequencies

BVP solver initiated and executed while using provided initial estimates, field equation function and boundary conditions function

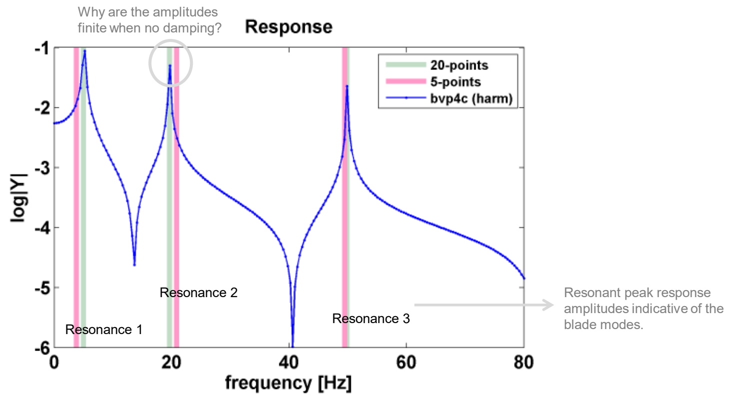
Field equation in the state space format (set of the first order ODEs)

Vector of the BC residuals (minimized to reach 0 and satisfy the BCs)

First solution estimate is 0, subsequent estimates are extrapolated from the previous runs.

Comparison of basic and bvp4c collocation solutions

Consider the harmonic BVP – rotating flapping beam – introduced previously. Compare two approximate solutions – one obtained using bvp4c and other obtained using basic (“naive”) collocation implementation. In the latter case, use the coarse (5 point) and fine (20 point) collocation meshes.



Summary

- Analytical eigenvalue BVP example shows that there is an infinite number of vibration modes.
- Approximate methods discretize 1D domain and establish the criteria for approximate solutions.
- MWR minimizes the weighted residuals of the field equations while BCs are always satisfied.
- Collocation methods satisfy the field equation residual requirements in the chosen collocation points.
- Bvp4c represents a numerically-robust collocation BVP solver.